CSE 413

Project Report

Simulation and Modeling Lab

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Introduction

This project analyzes whether completing a test preparation course has a significant impact on students' Math scores. The analysis is performed using a Two-Sample T-Test, supported by various statistical checks and visualizations. The dataset includes Math, Reading, and Writing scores along with demographic and preparation-related attributes.

The main goal is to statistically test:

Does completing the test preparation course result in higher Math scores compared to not completing it?

Brief Theory

Two-Sample T-Test

A two-sample t-test compares the means of two independent groups to see if they are significantly different.

- Null Hypothesis (H_0):
  $\mu_1 = \mu_2$ (No difference in means)
- Alternative Hypothesis (H_1):
  $\mu_1 \neq \mu_2$ (Means are different)

Levene's Test

Before running the t-test, we check if variances of the two groups are equal.

- If $p > 0.05$ → Variances equal → pooled t-test
- If $p < 0.05$ → Variances unequal → Welch's t-test

Features Used

Primary Feature

- math_score - response variable

Grouping Feature

- test_preparation_course (completed / none)

Additional Features:

- reading_score

- writing_score
- gender

Equations Used

T-Statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Degrees of Freedom (Pooled t-test)

$$df = n_1 + n_2 - 2$$

Confidence Interval

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2} \times SE$$

Levene's Test Statistic

$$H_0 : \sigma_1^2 = \sigma_2^2$$

Dataset Details

Dataset was mounted from Google Drive.

Source:
Kaggle Dataset: Students Performance in Exams
`https://www.kaggle.com/datasets/spscientist/students-performance-in-exams`

Dataset Size: 1000 rows
Columns:

- gender

- race/ethnicity
- parental_level_of_education
- lunch
- test_preparation_course
- math_score
- reading_score
- writing_score

Summary of Code Implementation

Data Extraction

```
group_completed = df[df['test_preparation_course'] ==
'completed']['math_score']
group_none = df[df['test_preparation_course'] == 'none']['math_score']
```

Variance Equality Check (Levene Test)

```
levene_stat, levene_p = stats.levene(group_completed, group_none)
```

Levene's Test p-value: 0.4655125071689348

Two-Sample T-Test

```
t_statistic, p_value = stats.ttest_ind(group_completed, group_none,
equal_var=True)
```

Results

- Mean (Completed): 69.70
- Mean (None): 64.08
- T-statistic: 5.70
- p-value: 1.53e-08
- 95% CI: [3.69, 7.55]
- df: 998

Graphs

Fig 1: Boxplot of Math Score by Test Prep

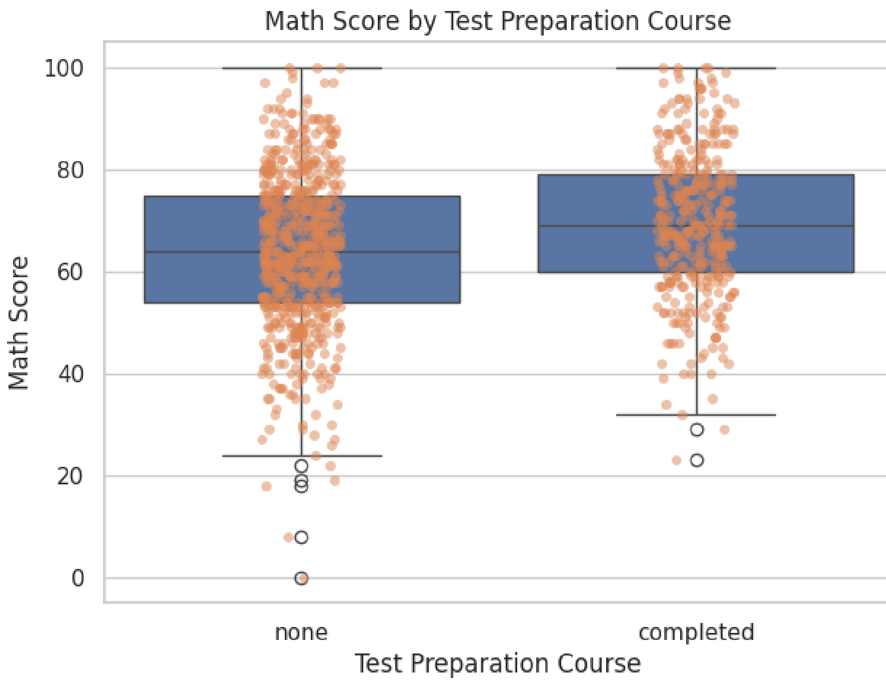


Fig 2: Distribution Comparison (Hist + KDE)

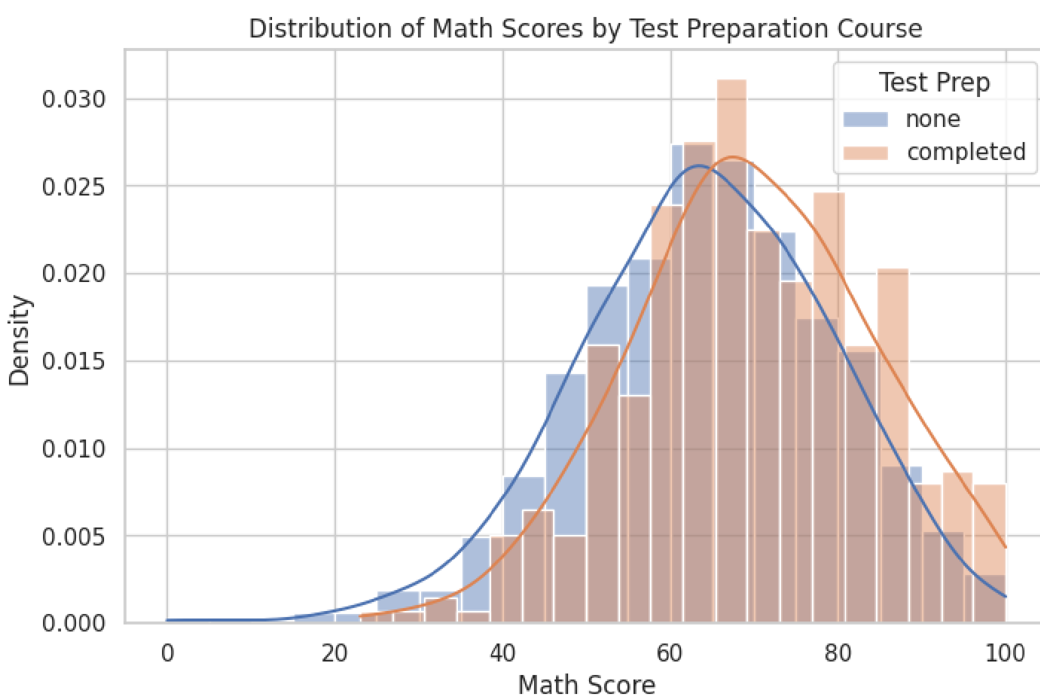


Fig 3: Violin Plot

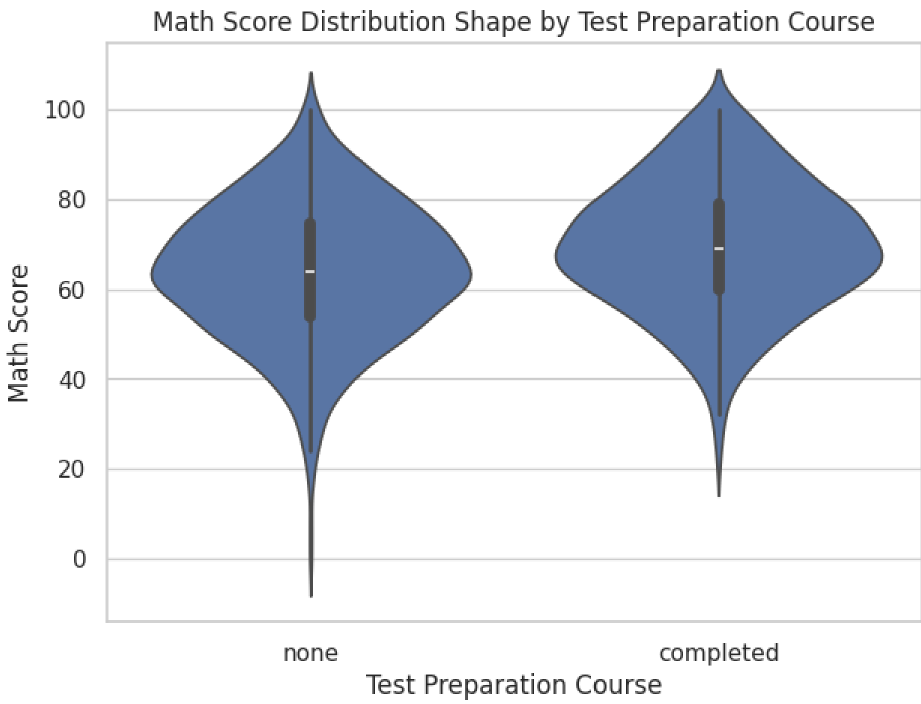


Fig 4: Mean ± 95% CI Plot

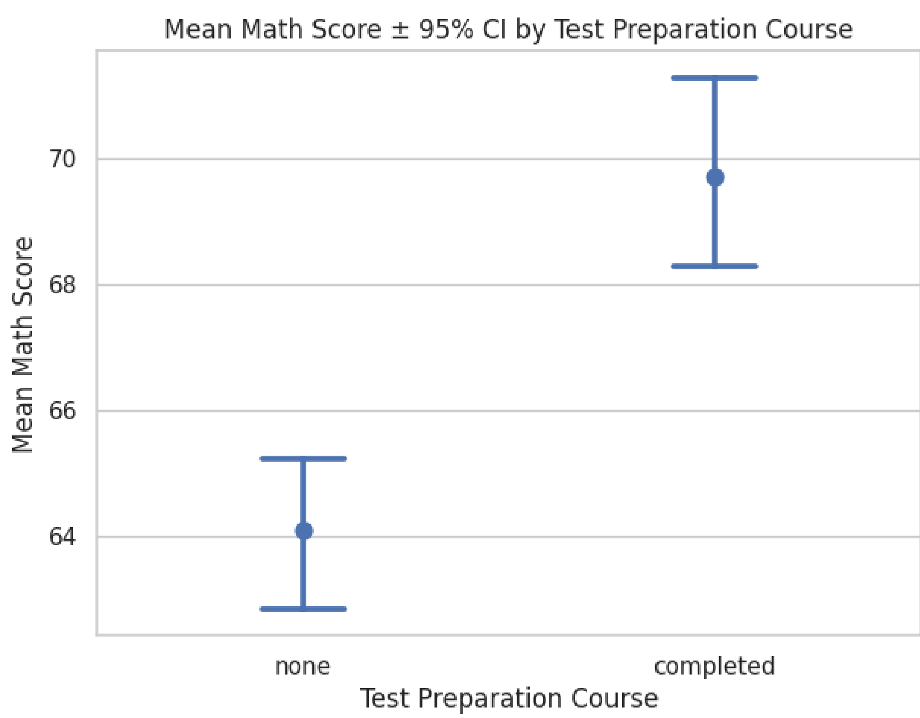


Fig 5: ECDF Curve

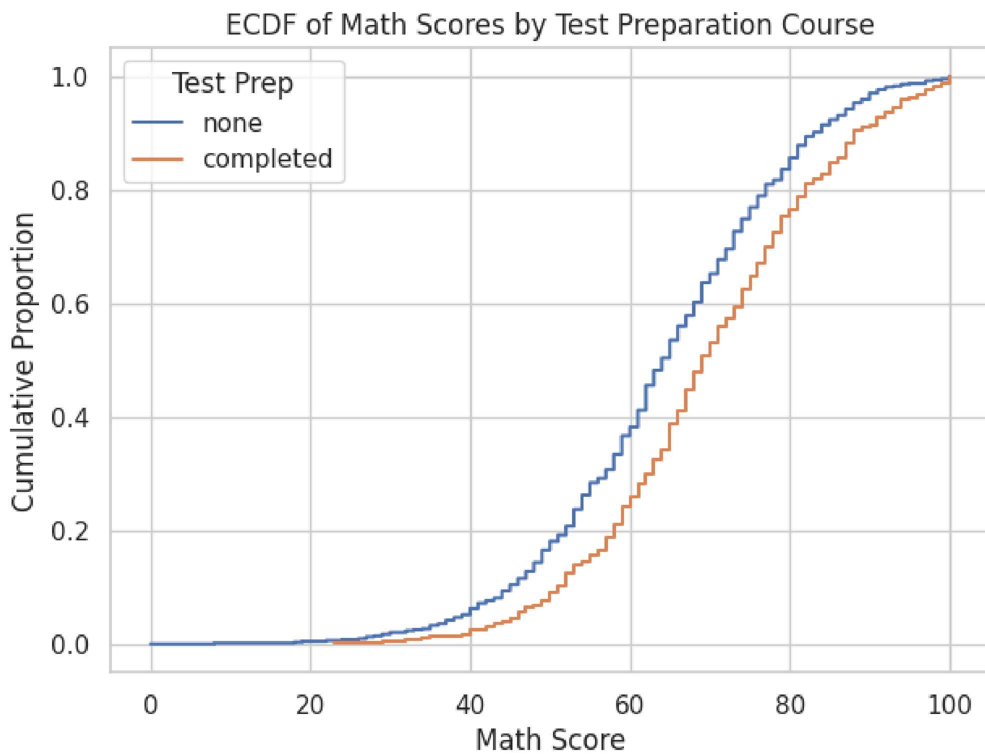


Fig 6: Gender-wise Comparison

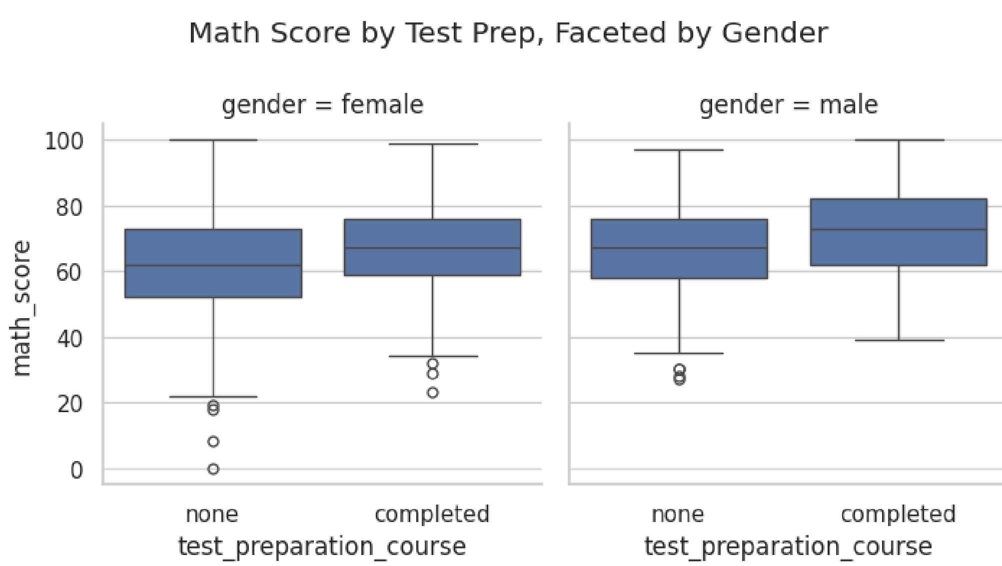


Fig 7: Math vs Reading Scatter

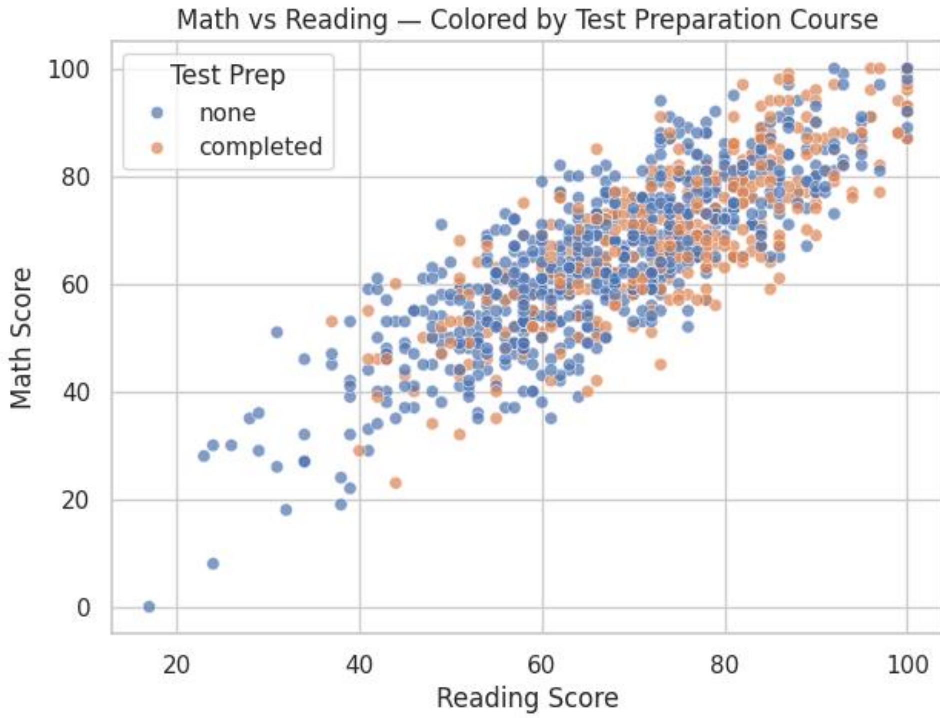


Fig 8: Regression Lines

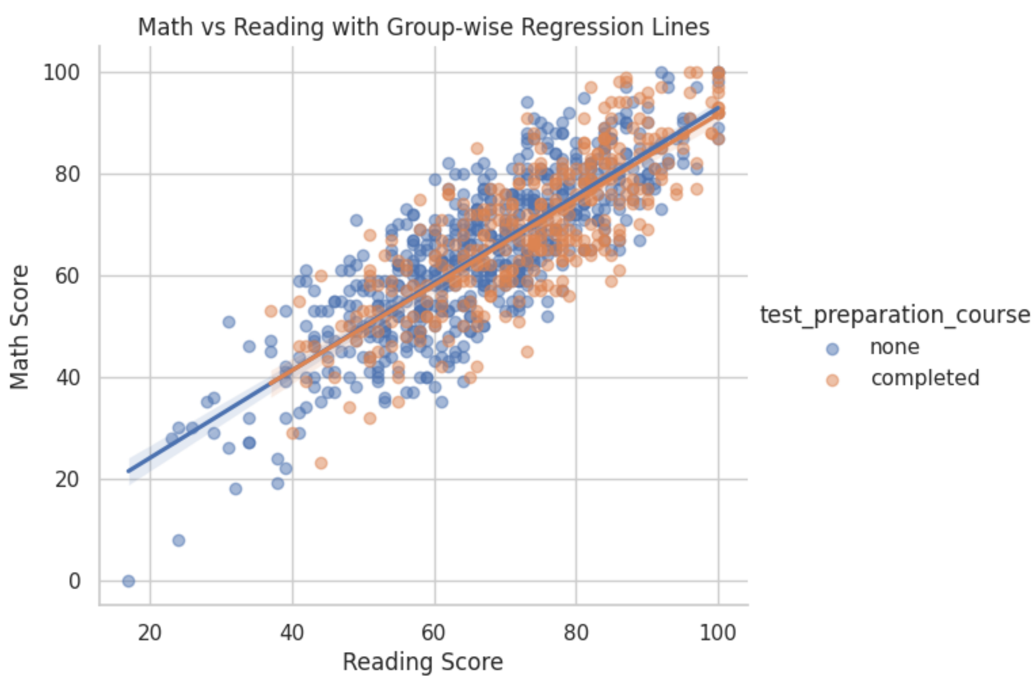


Fig 9: Pair plot

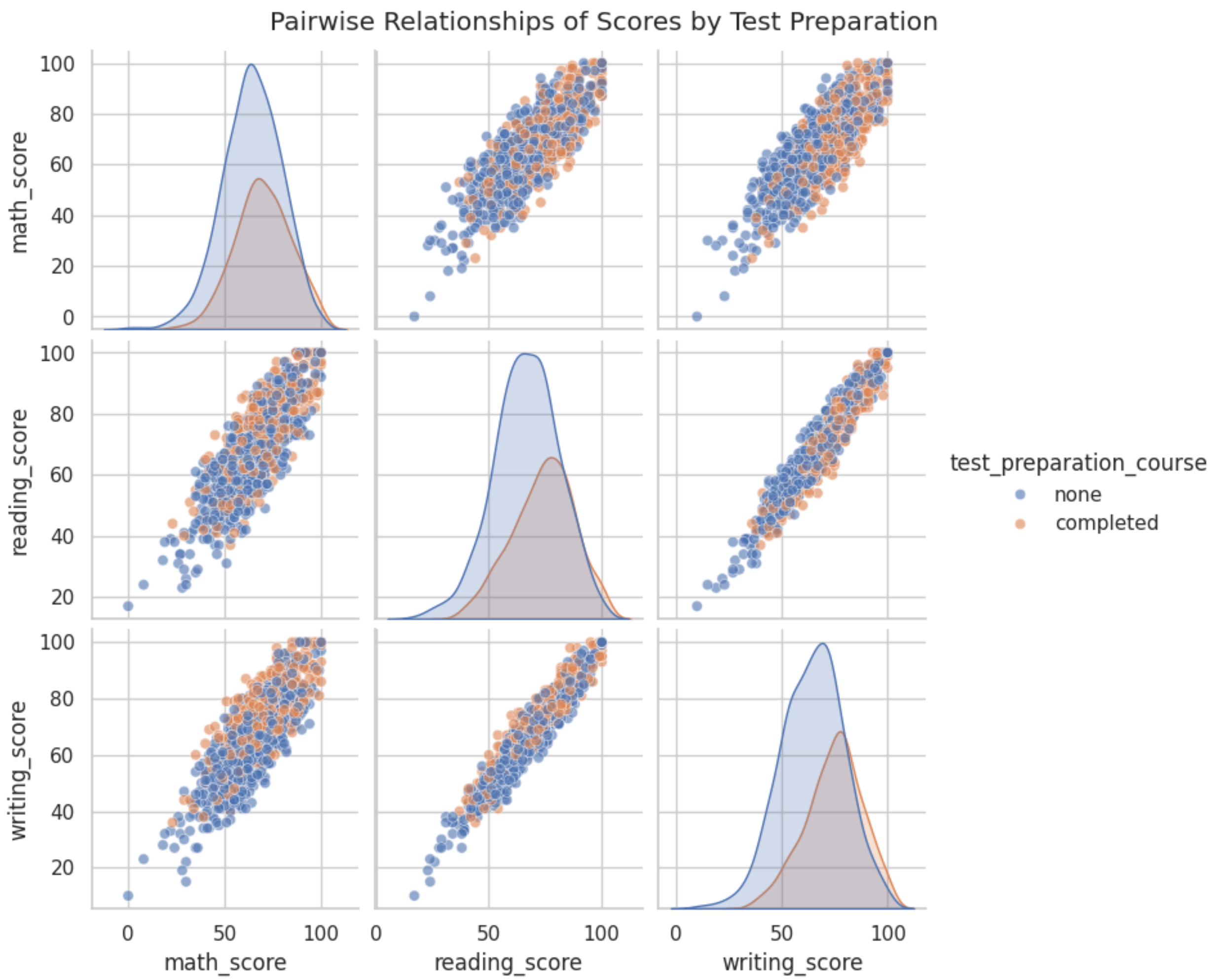


Fig 10: Correlation Heatmap

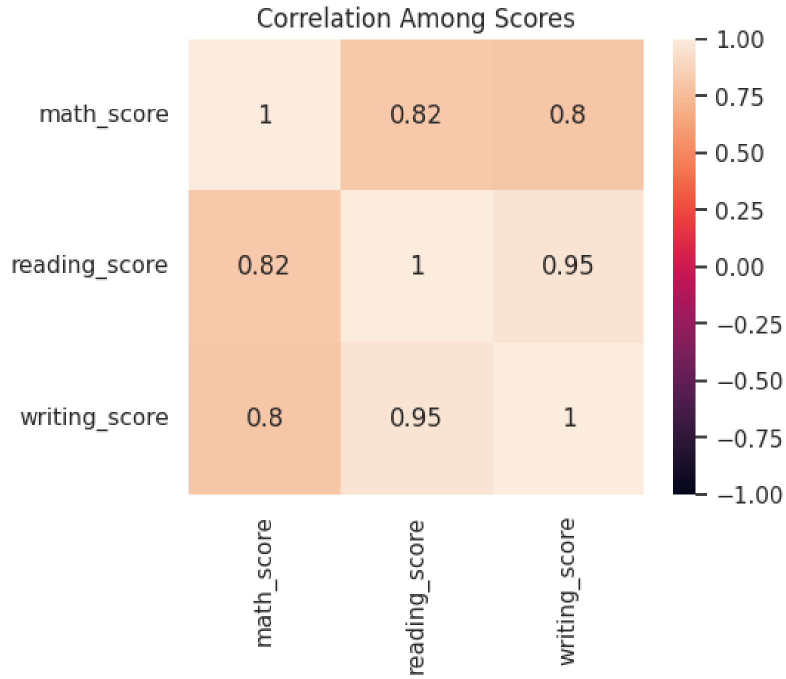


Fig 11: Subject-wise Boxplot

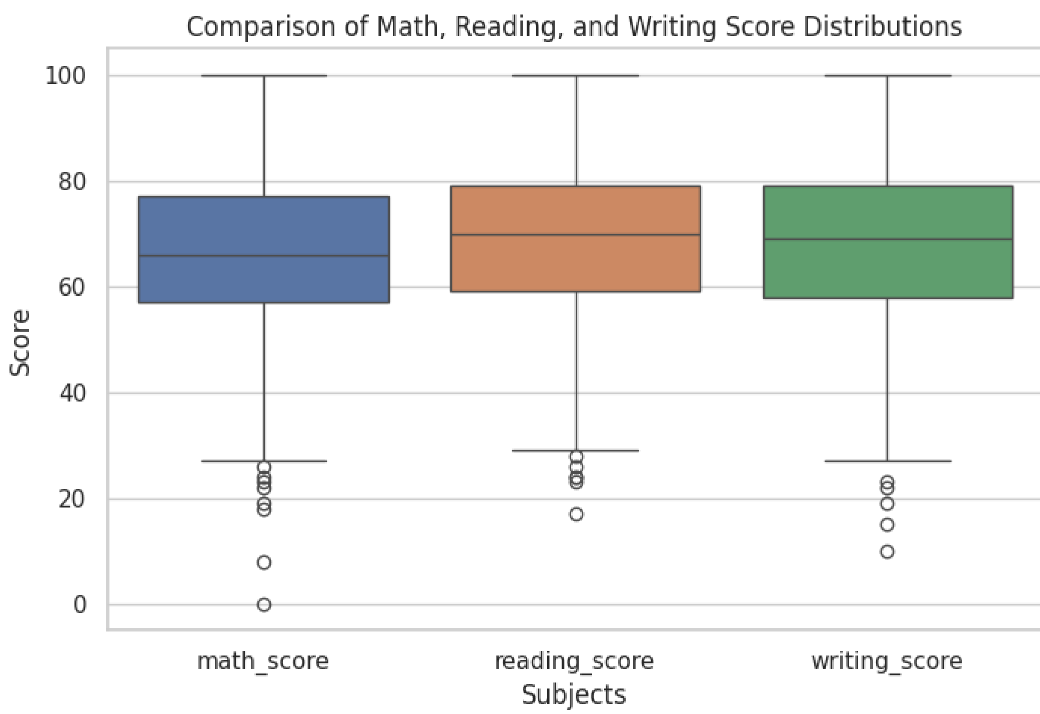
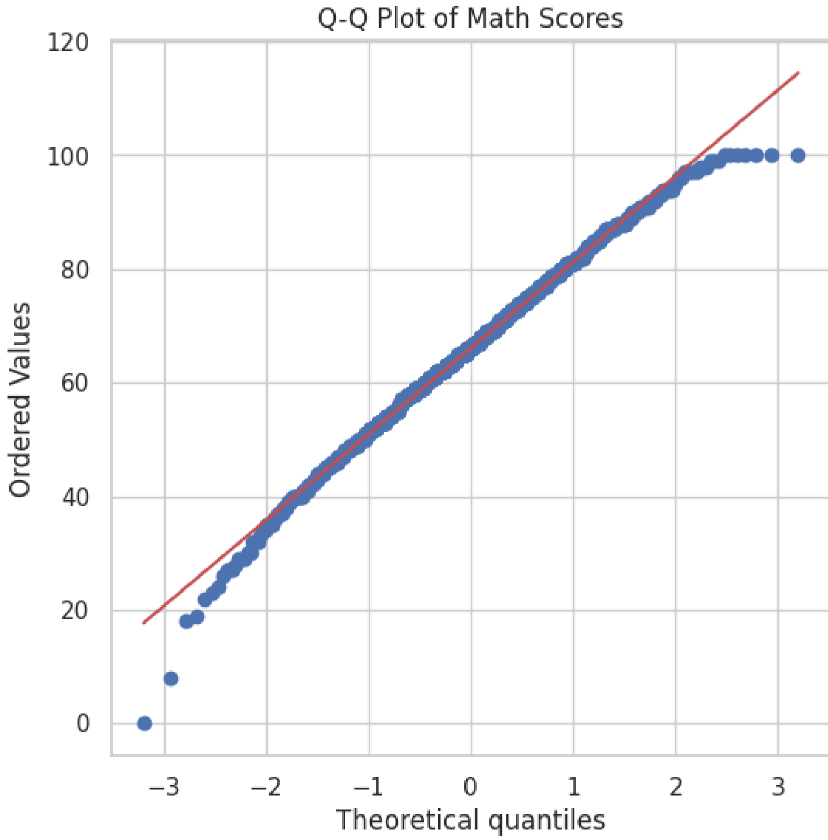


Fig 12: Q-Q Plot



Result Analysis

students who completed the test preparation course scored higher in Math compared to those who did not. The Levene's Test showed equal variances, that's why we used a pooled t-test. Both the statistical test and visualizations consistently show that the distribution of Math scores for the "completed" group is shifted upward.

So, the preparation course demonstrates a measurable and consistent improvement in Math performance.

GitHub Repository Link

Ramim19/Simulation-and-modeling-lab-Project-CSE-413: Impact of Test Preparation on Students' Math Performance: A Two-Sample T-Test with Data Visualization